Entanglement-Entropy Sourcing of Gravity: Dimensional Repair, Covariant Completion, and Experimental Outlook

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Abstract

We develop a dimensionally consistent phenomenological extension of thermodynamic gravity in which a *protocol-certified* internal entanglement entropy density of a quantum device acts as an additional gravitational source. The central repair over earlier drafts is twofold: a reference energy scale E_* (J/bit) converts dimensionless entanglement counts to an energy density, and the source field is redefined from an ambiguous "internal entropy" to $S_{\mathrm{cert}}(x)$ — entanglement generated by a specified control sequence, certified by a witness, and varied between experimental branches. This operational definition, grounded in the theory of accessible entanglement under superselection rules, excludes uncontrolled background correlations. The Cooper-pair constraint is resolved not by asserting that BCS entanglement is absent — it is not — but by a **Cancellation Proposition**: because the substrate state is identical in both experimental branches, background entanglement cancels exactly in the differential observable ΔS_{cert}. An explicit open conjecture (OP-new) is stated for the residual contributions. Under the Planck-scale benchmark $E_* = E_P$, all observables reduce to $\Lambda_{\mathrm{eff}} = \tilde{\kappa} E_* \alpha_{\mathrm{screen}}$, with a predicted unsuppressed acceleration of $\Delta a = 3.6 \times 10^{-9}\,\mathrm{m/s^2}$ for 100 protocol-certified entangled bits at $100\,\mu$m, within reach of cryogenic SQUID-based force sensors.

1 Purpose and Scope

This document is a dimensional-consistency revision of the entanglement–gravity program. Its goal is not to select a unique microscopic origin for the coupling, but to isolate the minimal scaling assumptions required for a well-posed phenomenological theory.

Earlier drafts used a coupling prefactor proportional to $c^4/(8\pi G k_B \ln 2)$ multiplying an entanglement entropy density measured in bits per unit volume. In SI units, that expression does not yield an energy density, so the resulting stress–energy tensor and derived accelerations were not dimensionally valid. The present revision fixes that problem by introducing a reference energy scale E_* and rebuilding the source term around it.

2 Operational Definition and Units

2.1 The dimensional problem

The Einstein equations

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{1}$$

require $[T_{\mu\nu}] = \mathrm{J\,m^{-3}}$. Writing $[S_{\mathrm{ent}}] = \mathrm{bit\,m^{-3}}$ (bit dimensionless) and using the original coupling $\alpha = \tilde{\kappa} c^4/(8\pi G k_B \ln 2)$ gives $[\alpha S_{\mathrm{ent}}] = \mathrm{K\,m^{-4}}$, not an energy density. A reference energy scale E_* (J/bit) is the minimal repair.

2.2 Protocol-certified entanglement: S_{cert}

The earlier notation S_{cert} was ambiguous: it could be interpreted as the von Neumann entropy of any reduced density matrix, including thermal and BCS correlations in the device substrate. We replace it with a precisely defined operational quantity.

Definition 2.1 (Protocol-certified entanglement density)**.** Let D be a quantum device, P a specified gate sequence acting on the device degrees of freedom, $D = A \cup B$ a bipartition pre-registered before P is applied, and $\mathcal{W}$ an entanglement witness verified after P. The **protocol-certified entanglement density** is

$$S_{\text{cert}}(x) = \frac{-k_B \operatorname{Tr}_A\left[\rho_A^P(x) \ln \rho_A^P(x)\right]}{k_B \ln 2\, \Delta V}, \quad \rho_A^P = \operatorname{Tr}_B\left[P \rho_D P^\dagger\right], \tag{2}$$

where $P\rho_D P^\dagger$ is the device state after protocol P and ΔV is the cell volume. We require that $\mathcal{W}$ certifies $S_{\text{cert}} > 0$ to within experimental error before a run is accepted.

This replaces the generic bipartite von Neumann entropy used in earlier drafts. The definition has three operationally necessary properties: (i) $S_{\text{cert}} \geq 0$; (ii) $S_{\text{cert}} = 0$ if no entangling protocol was applied; (iii) S_{cert} is certified by a positive entanglement witness, so it cannot be mimicked by classical correlations.

3 Minimal Dimensional Fix

3.1 Reference energy scale

We introduce a reference energy scale E_* with units of energy per bit and define the entanglement energy density by

$$\rho_{\text{ent}}(x) = E_*\, S_{\text{ent}}(x). \tag{3}$$

Then

$$[\rho_{\text{ent}}] = [E_*][S_{\text{ent}}] = \mathrm{J\,m^{-3}}, \tag{4}$$

as required.

An equivalent parameterization is

$$\rho_{\text{ent}}(x) = \lambda_* k_B T_* s_{\text{ent}}(x), \tag{5}$$

where s_{ent} is entropy density in thermodynamic units, T_* is an effective temperature scale, and λ_* is dimensionless. In this white paper we use the simpler single-scale notation E_*.

3.2 Revised source term

The entanglement contribution to the stress–energy tensor is rewritten as

$$T_{\mu\nu}^{\text{ent}}(x) = \tilde{\kappa}\, \rho_{\text{ent}}(x) g_{\mu\nu} + (\text{gradient terms}), \tag{6}$$

with

$$\rho_{\text{ent}}(x) = E_* S_{\text{ent}}(x). \tag{7}$$

Thus all previous appearances of αS_{ent} are replaced by $\tilde{\kappa} E_* S_{\text{ent}}$.

If desired, one may later relate E_* to a microscopic scale such as

$$E_* = \eta k_B T_*, \tag{8}$$

with η dimensionless and T_* an effective Unruh or horizon temperature. The present framework leaves the origin of E_* open.

4 Revised Effective Action and Fifth-Force Equation

4.1 External-source action

Let

$$\phi(x) \equiv S_{\text{ent}}(x). \tag{9}$$

We retain the external-source interpretation: ϕ is prescribed by the device quantum state and is not varied as an independent field. The entanglement-sector Lagrangian density is written as

$$\mathcal{L}_{\text{ent}} = -\frac{\beta}{2} g^{\mu\nu}(\nabla_\mu \phi)(\nabla_\nu \phi) - \lambda\phi, \tag{10}$$

with

$$\lambda = \tilde{\kappa} E_*. \tag{11}$$

The full action is therefore

$$S[g;\phi] = \int d^4x \sqrt{-g} \left[\frac{c^4}{16\pi G} R - \frac{\beta}{2} g^{\mu\nu}(\nabla_\mu \phi)(\nabla_\nu \phi) - \tilde{\kappa} E_* \phi \right]. \tag{12}$$

Varying with respect to $g_{\mu\nu}$ yields

$$T^{\text{ent}}_{\mu\nu} = \beta \left(\nabla_\mu \phi \, \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} (\nabla\phi)^2 \right) + \tilde{\kappa} E_* \phi \, g_{\mu\nu}. \tag{13}$$

The dimensions are correct provided β carries the dimensions needed to make the gradient term an energy density. In slowly varying configurations, the gradient term is subleading.

4.2 Matter nonconservation

Total conservation requires

$$\nabla^\mu T^{\text{total}}_{\mu\nu} = 0, \tag{14}$$

so that

$$\nabla^\mu T^{\text{matter}}_{\mu\nu} = -\nabla^\mu T^{\text{ent}}_{\mu\nu}. \tag{15}$$

Substituting the revised entanglement tensor gives

$$\nabla^\mu T^{\text{matter}}_{\mu\nu} = -\left(\tilde{\kappa} E_* + \beta \Box \phi\right) \nabla_\nu \phi + \beta \nabla_\nu \left(\frac{1}{2} (\nabla\phi)^2 \right). \tag{16}$$

In the quasi-static regime with modest gradients and $|\beta \Box \phi| \ll |\tilde{\kappa} E_*|$, this reduces to

$$\nabla^\mu T^{\text{matter}}_{\mu\nu} \approx -\tilde{\kappa} E_* \, \nabla_\nu \phi(x) = -\tilde{\kappa} E_* \, \nabla_\nu S_{\text{ent}}(x). \tag{17}$$

This is the corrected fifth-force law. Its units are now those of force density, as required.

5 Newtonian Limit and Revised Observable

5.1 Modified Poisson equation

In the Newtonian limit,

$$\nabla^2\Phi = 4\pi G\left(\rho_{\rm matter} + \rho_{\rm ent}\right), \tag{18}$$

with

$$\rho_{\rm ent}(\mathbf{x}) = \tilde{\kappa} E_* S_{\rm ent}(\mathbf{x}). \tag{19}$$

Under spherical symmetry,

$$\Delta a(R) = -\frac{d\Phi_{\rm ent}}{dR} = \frac{G}{R^2} M_{\rm ent}(< R), \tag{20}$$

where

$$M_{\rm ent}(< R) = 4\pi \int_0^R \rho_{\rm ent}(r) r^2\, dr. \tag{21}$$

For bounded sources, $M_{\rm ent}(< R)$ saturates outside the source and the correction scales as $1/R^2$.

5.2 Observable combination

Experiments constrain the product

$$\Lambda_{\rm eff} \equiv \tilde{\kappa}\, E_*\, \alpha_{\rm screen}, \quad [\Lambda_{\rm eff}] = \mathrm{J/bit}, \tag{22}$$

not $\tilde{\kappa}$ or E_* individually. In branch-comparison experiments the useful observational combination is

$$\mathcal{B} \equiv |\Lambda_{\rm eff}|\, \frac{\Delta S_{\rm ent}^{\rm tot}}{R_{\rm eff}^2}, \quad [\mathcal{B}] = \mathrm{J\,m^{-3}}, \tag{23}$$

where $\Delta S_{\rm ent}^{\rm tot}$ is the differential internal entanglement content between experimental branches and $R_{\rm eff}$ is the characteristic geometric scale. A null result at sensitivity $\Delta a_{\rm max}$ constrains

$$|\Lambda_{\rm eff}| < \frac{\Delta a_{\rm max}\, R_{\rm eff}^2}{G\, \Delta S_{\rm ent}^{\rm tot}}. \tag{24}$$

Mapping this bound to $|\tilde{\kappa}|$ alone requires an independent assumption about E_*; without such an assumption, only $|\Lambda_{\rm eff}|$ is constrained.

6 Implications for OP8 and OP10

6.1 Fifth-force bounds

The corrected acceleration outside a bounded source of radius R_0 is

$$\Delta a(R) = \frac{G\, |\Lambda_{\rm eff}|\, \Delta S_{\rm ent}^{\rm tot}}{R^2}, \quad R > R_0. \tag{25}$$

A null force bound $\Delta a_{\rm max}$ constrains the observable product

$$|\Lambda_{\rm eff}| = |\tilde{\kappa}|\, E_*\, \alpha_{\rm screen} \lesssim \frac{\Delta a_{\rm max}\, R_{\rm eff}^2}{G\, \Delta S_{\rm ent}^{\rm tot}}. \tag{26}$$

Benchmark (unsuppressed signal). For the Planck-scale hypothesis $E_* = E_P = 1.96 \times 10^9\,\mathrm{J}$ and $|\tilde{\kappa}| = 1/4$, the effective mass per bit is $m_{\mathrm{bit}} = |\tilde{\kappa}| E_P/c^2 = 5.44\,\mu\mathrm{g/bit}$. For $\Delta S_{\mathrm{ent}} = 100\,\mathrm{bits}$ at $R = 100\,\mu\mathrm{m}$ with $\alpha_{\mathrm{screen}} = 1$:

$$\Delta a = \frac{G \times 100 \times 5.44 \times 10^{-9}\,\mathrm{kg}}{(10^{-4}\,\mathrm{m})^2} = 3.6 \times 10^{-9}\,\mathrm{m/s^2}. \tag{27}$$

Cryogenic SQUID-based force sensors achieve $\sim 10^{-10}\,\mathrm{m/s^2}/\sqrt{\mathrm{Hz}}$, placing this signal above the noise floor by a factor of $\sim$36 at 1 Hz bandwidth.

Critical constraint. If $E_* \approx E_P$, macroscopic superconductors containing $\sim 10^{20}$ Cooper pairs would carry an effective entanglement mass of order 10^{11} kg per gram of material, producing enormous gravitational anomalies not observed. The framework requires that the operational definition of S_{cert} excludes uncontrolled thermal correlations (including BCS Cooper-pair entanglement) for which no controlled bipartition protocol was applied. Verifying this claim rigorously is an open problem; see Section 7.

6.2 RG interpretation

With E_* explicit, the natural renormalization-group ansatz becomes

$$\tilde{\kappa}_{\mathrm{lab}} E_{*,\mathrm{lab}} = \tilde{\kappa}_{\mathrm{UV}} E_{*,\mathrm{UV}} \left(\frac{E_{\mathrm{lab}}}{E_P}\right)^{\gamma}, \tag{28}$$

where E_P is the Planck energy and γ is an effective scaling exponent. Experiments therefore constrain the infrared product $\tilde{\kappa} E_*$ rather than $\tilde{\kappa}$ alone.

7 Operational Sourcing, Cancellation, and Open Problems

7.1 Why S_{cert} is the right object

The definition of S_{cert} in Section 2.2 may appear restrictive. This subsection explains why it is the *only* definition consistent with the goals of the framework.

The gravitational source must satisfy three physical requirements:

(i) **Branch-modulation:** it must differ between the high-entanglement branch A and the reference branch B, otherwise no differential signal exists.
(ii) **Controllability:** it must be adjustable by the experimenter, not fixed by the material.
(iii) **Certifiability:** it must be verifiable that entanglement, not classical correlations, is responsible for any signal.

The von Neumann entropy of an arbitrary reduced density matrix fails (i): BCS ground states, thermal states of any system in equilibrium, and nuclear spins all contribute a background S_{vN} that is fixed by the material and identical in both branches. S_{cert} satisfies all three requirements by construction.

This is not merely a convenient definition — it is the operationally accessible entanglement in the sense of Wiseman and Vaccaro [3]: entanglement that can be extracted and used as a resource under the device's allowed operations (local unitary gates on the qubit register, projective readout, classical communication). Background correlations that are not modifiable by the protocol are not resources under this definition, and are therefore not sources in this framework.

7.2 Cancellation theorem for background correlations

Proposition 7.1 (Background cancellation)**.** *Let the total system be $T = D \cup B_{\mathrm{g}}$, where D is the quantum device and B_{g} is the background (substrate, Cooper pairs, thermal phonons). Assume:*

(i) ***Product initial state:*** $\rho_T^{(0)} = \rho_D^{(0)} \otimes \rho_{B_{\mathrm{g}}}^{(0)}$.

(ii) ***Protocol locality:*** *the branch protocol acts on D only: $P^{A,B} = U_D^{A,B} \otimes \mathbf{I}_{B_{\mathrm{g}}}$.*

(iii) ***Bipartition locality:*** *the bipartition $\mathcal{A}$ is pre-registered on D only, so the witness $\mathcal{W}$ measures correlations within D.*

Then the differential protocol-certified entanglement of the background vanishes exactly:

$$\Delta S_{\mathrm{cert}}^{B_{\mathrm{g}}} \equiv S_{\mathrm{cert}}\big(\rho_{B_{\mathrm{g}}}; P^A, \mathcal{A}\big) - S_{\mathrm{cert}}\big(\rho_{B_{\mathrm{g}}}; P^B, \mathcal{A}\big) = 0. \tag{29}$$

The measured signal depends only on $\Delta S_{\mathrm{cert}}^D = S_{\mathrm{cert}}^A - S_{\mathrm{cert}}^B$ for the device qubits.

Proof. Under assumption (ii), $P^{A,B}$ acts as the identity on B_{g}. Therefore $P^{A,B}\rho_{B_{\mathrm{g}}}P^{A,B\dagger} = \rho_{B_{\mathrm{g}}}$ for both branches. Under assumption (iii), the witness $\mathcal{W}$ measures correlations across the bipartition $\mathcal{A}$ of D; it has no action on B_{g}. Consequently $S_{\mathrm{cert}}(\rho_{B_{\mathrm{g}}}; P^A) = S_{\mathrm{cert}}(\rho_{B_{\mathrm{g}}}; P^B) = 0$, and Eq. (29) follows. □

Physical meaning. The Cooper-pair constraint is resolved not by claiming BCS entanglement is absent — it is not — but by the observation that BCS correlations are *branch-independent*. The substrate state $\rho_{B_{\mathrm{g}}}$ is identical in branches A and B: both branches run the same physical setup (same temperature, same chip, same equilibrium state) with the only difference being the CZ gate phase applied to the qubit register. A state that is identical in both branches contributes zero to the differential observable ΔS_{cert} and zero to the gravitational signal.

Validity of assumption (i). For superconducting transmon qubits, the device-substrate coupling is phononic ($J_{\mathrm{ph}} \lesssim 1\,\mathrm{kHz}$) relative to qubit transition frequencies ($\omega/2\pi \sim 5\,\mathrm{GHz}$), giving a ratio $J_{\mathrm{ph}}/\omega \sim 10^{-7}$. The initial state is therefore a product state to excellent approximation. Gate-induced phonon backaction creates device-substrate entanglement as a *systematic*, which must be matched between branches (identical gate sequences on branch B); it does not constitute S_{cert} under Definition **??**.

7.3 OP-new: thermal equilibrium and S_{cert}

Proposition 7.1 resolves the Cooper-pair constraint for differential observables. A residual question remains about the *absolute* gravitational source from any system in thermal equilibrium, independent of branch comparison.

Conjecture 7.1 (OP-new: Thermal state conjecture)**.** For any quantum system in a Gibbs state $\rho_\beta \propto e^{-\beta H}$, subject to the operational constraints (local controls, charge/particle-number superselection, fixed bipartition) of a realistic laboratory device, the protocol-certified entanglement satisfies

$$S_{\mathrm{cert}}(\rho_\beta; P_{\mathrm{id}}, \mathcal{A}, \mathcal{W}) = 0, \tag{30}$$

where P_{id} is the identity protocol (no entangling unitary applied).

Why this conjecture is plausible. When $P = P_{\mathrm{id}}$, no entanglement is *generated* by the protocol. The witness $\mathcal{W}$ then measures the entanglement already present in the initial state under the allowed operations. For a Gibbs state at finite temperature, Conjecture 7.1 amounts to asking whether thermal equilibrium states have operationally accessible entanglement (in the Wiseman-Vaccaro sense) under realistic device operations.

For BCS superconductors specifically, there is a further suppression from the charge superselection rule: under strict U(1) charge conservation, Cooper-pair entanglement across a spatial cut is not operationally accessible (Verstraete-Cirac, 2003 [4]). This provides partial support for the conjecture in the superconducting case.

Proof strategy for OP-new. Three layers of evidence are needed:

1. *Free-fermion model.* Compute $S_{\mathrm{cert}}(\rho_\beta)$ for a 1D free-fermion chain at temperature T under the constraint that the protocol P consists only of single-site unitaries (no entangling gates). Prediction: $S_{\mathrm{cert}} = 0$ because no entanglement can be generated by single-site operations from a thermal product state.
2. *BCS-like model.* Compute $S_{\mathrm{cert}}(\rho_{\mathrm{BCS}})$ for the reduced state of a spatial half-chain under the charge SSR. Prediction: $S_{\mathrm{cert}} \leq E_{\mathcal{P}}^{\mathrm{SSR}}$ (accessible entanglement under SSR), which vanishes in the strict-SSR limit [3].
3. *General argument.* Use the monotonicity of relative entropy under CPTP maps to show that the accessible entanglement of ρ_β under the device's operational restrictions is bounded by a quantity that vanishes as $T \to \infty$ and as the SSR becomes strict.

A negative resolution of Conjecture 7.1 — showing that equilibrium S_{cert} is large under realistic operations — would require either $E_* \ll E_P$ (rendering all near-term experiments insensitive) or a fundamental reformulation of the sourcing mechanism.

7.4 Status of E_* and the constrained combination

The reference energy E_* is an EFT matching coefficient that the framework does not determine. Only the product

$$\Lambda_{\mathrm{eff}} \equiv \tilde{\kappa}\, E_*\, \alpha_{\mathrm{screen}} \quad [\mathrm{J/bit}] \tag{31}$$

is experimentally constrained. The three physically motivated choices span many orders of magnitude:

Choice	E_*	Δa (100 bits, 100 μm, $\alpha_{\mathrm{screen}} = 1$)
Planck scale ($E_* = E_P$)	1.96×10^9 J	3.6×10^{-9} m/s^2
Qubit energy ($E_* = \hbar\omega$, 5 GHz)	3.3×10^{-24} J	6×10^{-42} m/s^2
Thermal ($E_* = k_B T$, 15 mK)	2.1×10^{-25} J	4×10^{-43} m/s^2

Only the Planck-scale choice produces a signal above any conceivable near-term threshold, making it the benchmark hypothesis. This sharpens the importance of OP-new (does thermal S_{cert} vanish?), OP3 (what is α_{screen} for realistic NISQ circuits?), and OP10 (does $\tilde{\kappa}E_*$ renormalize from the Planck scale to the laboratory?).

8 Conclusions

The entanglement–gravity framework has reached a state of internal consistency. Three points define its honest status:

1. The source is operationally defined. The gravitational source field $S_{\text{cert}}(x)$ is protocol-certified entanglement density: entanglement generated by a specified control sequence, certified by an entanglement witness relative to a pre-registered bipartition, and varied between experimental branches. Background many-body correlations — including BCS Cooper-pair entanglement and all thermal correlations — are excluded by the Cancellation Proposition 7.1: they are branch-independent and cancel exactly in the differential observable.

2. The observable is $\Lambda_{\text{eff}} = \tilde{\kappa} E_* \alpha_{\text{screen}}$. Neither $\tilde{\kappa}$, E_*, nor α_{screen} is individually constrained by any experiment. Only their product is. Mapping a null result to a bound on $\tilde{\kappa}$ requires independent input from OP3 (Lindblad numerics for α_{screen}) and OP10 (RG flow of $\tilde{\kappa} E_*$ from UV to laboratory scales).

3. Force sensing is the correct experimental path. For $E_* = E_P$, $|\tilde{\kappa}| = 1/4$, $\Delta S_{\text{cert}} = 100\,\text{bits}$, $R = 100\,\mu\text{m}$, the predicted unsuppressed acceleration is $\Delta a = 3.6 \times 10^{-9}\,\text{m/s}^2$. Cryogenic SQUID-based force sensors achieve $\sim 10^{-10}\,\text{m/s}^2/\sqrt{\text{Hz}}$, placing the signal above the noise floor by a factor of ~ 36. Ramsey-interferometry phase shifts are suppressed by c^2 and lie far below current thresholds.

The framework is now *falsifiable and tightly constrained*: a well-designed differential force-sensing experiment, combined with theoretical progress on OP-new (Conjecture 7.1), OP3, and OP10, will either detect an entanglement-dependent gravitational anomaly or rule it out at the Planck-scale benchmark level.

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